

Source: Stewart, Section 2.3 pp 94 - 104

Limit Laws

Suppose that c is a constant and the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist. Then

$$1 \quad \lim_{x \rightarrow a} [f(x) + g(x)] =$$

$$\text{Ex: } \lim_{x \rightarrow 2} (x^2 + x) =$$

$$2 \quad \lim_{x \rightarrow a} [f(x) - g(x)] =$$

$$\text{Ex: } \lim_{x \rightarrow 2} (x^2 - x) =$$

$$3 \quad \lim_{x \rightarrow a} [c f(x)] =$$

$$\text{Ex } \lim_{x \rightarrow 2} 4x =$$

$$4 \quad \lim_{x \rightarrow a} [f(x) g(x)] =$$

$$\text{Ex: } \lim_{x \rightarrow \frac{\pi}{6}} \sin x \cdot \cos x =$$

$$5 \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} =$$

$$6 \quad \lim_{x \rightarrow a} [f(x)]^n =$$

$$7 \quad \lim_{x \rightarrow a} \sqrt[n]{f(x)} =$$

$$8 \quad \lim_{x \rightarrow a} c =$$

$$9 \quad \lim_{x \rightarrow a} x =$$

$$10 \quad \lim_{x \rightarrow a} x^n =$$

$$11 \quad \lim_{x \rightarrow a} \sqrt[n]{x} =$$

EXAMPLES: Find

$$1. \quad \lim_{x \rightarrow 2} (3x^2 + 4 \sin \frac{\pi}{x} + 5)$$

$$2. \quad \lim_{x \rightarrow 0} \frac{x}{\sqrt{x+4} - 2}$$

The Squeeze Theorem

If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as $x \rightarrow a$, then



The Squeeze Theorem

If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

EXAMPLE:

$$\text{Find } \lim_{x \rightarrow 0^+} \sqrt{x} \sin \frac{1}{x}$$